

Hierarchical 3-D Scene-Graph-Based Semantic-Metric SLAM for Plant Inspection and Fruit Counting in Intelligent Hydroponics System

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Abstract—The synergistic development of Internet of Things (IoT), robotics, and artificial intelligence (AI) is reshaping the technological paradigms of interdisciplinary laboratories and industrial ecosystems. IoT-enabled vertical farming systems demonstrate significant advantages, achieving yield enhancement while reducing carbon emissions compared to traditional agriculture, thereby providing innovative solutions for sustainable food production. The advancement of robotic technologies further expands the application dimensions of mobile intelligent sensors in vertical farm IoT networks. Based on an autonomous farming system that integrates uncrewed aerial vehicle (UAV), sensors, and modular vertical farming units, this study proposes a 3-D scene graph (3DSG)-based hierarchical mapping method for the dynamic monitoring of plant and fruit growth. Through feedback mechanisms, the system optimizes growth conditions by adjusting lighting and nutrient delivery, while the hierarchical mapping architecture reduces detection errors and enables comprehensive 3-D visualization. The main contributions of this research include: 1) Pioneering application of 3DSG technology to establish a multidimensional spatiotemporal representation model for plant growth processes, supporting interpretable analysis and traceable monitoring; 2) Establishing an uncertainty model through error propagation by systematically analyzing sensor models (covering various common sensor combinations) and integrating these models into 3-D object pose estimation algorithms. This highlights the necessity of hierarchical abstraction levels. The system is validated through simulations and real-world experiments, providing a quantitative evaluation of object pose estimation; and 3) An IoT-driven intelligent vertical farming architecture that integrating mobile robotic perception networks and environmental regulation devices, enabling dynamic acquisition and closed-loop control of plant growth parameters. Open-source code is available at https://github.com/allenthree/scene_graph, video link: <https://youtu.be/dhc8RLmX7hc>.

Index Terms—Hierarchical scene graph, hydroponics platform, Internet of Things (IoT), metric-semantic SLAM, sustainable agriculture robots.

I. INTRODUCTION

UNDER the dual pressures of current population growth and climate change, food security has become an especially critical challenge for high-density areas like Hong Kong and Singapore. Smart agriculture, empowered by a multimodal technology cluster integrating large language model (LLM), robotics, and the Internet of Things (IoT), is fundamentally reshaping agricultural production systems. Through the 4-D synergy of perception, analysis, decision-making, and execution, a closed-loop system is established, encompassing data collection, intelligent decision-making, precise execution, and feedback evaluation. This innovation not only overcomes the resource efficiency bottlenecks inherent in traditional agriculture but also establishes a scientific paradigm that supports sustainable development. Driven by this new ecosystem of scientific instrumentation, this transformation aligns agricultural and food industries with the United Nations Sustainable Development Goals [1], offering a technological pathway to resolve food security dilemmas in land and resource-constrained regions.

To address these imperatives, we have initiated research on autonomous intelligent vertical farming. Vertical farming addresses critical challenges in densely populated regions (e.g., Hong Kong, Singapore), where arable land is scarce but high-rise buildings and vacant industrial facilities are abundant. Researchers have developed an integrated autonomous intelligent vertical farm system combining uncrewed aerial vehicles (UAVs) with vertical farming infrastructure [2]. These systems autonomously regulate light environment parameters and nutrient delivery strategies through perceptual feedback data, thereby forming an optimized closed-loop for plant growth. Our work aims to overcome the inherent limitations of traditional agriculture in terms of resource efficiency and environmental adaptability, while simultaneously constructing a closed-loop scientific discovery framework for next-generation smart agriculture.

In recent years, UAV technology has profoundly penetrated various sectors, including surveillance patrols, logistics transportation [3], and geospatial mapping [4]. At the intersection of IoT and smart agriculture, UAVs are evolving

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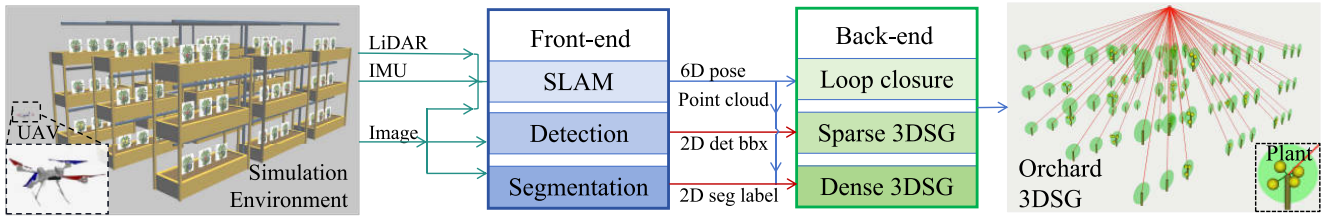


Fig. 1. System architecture of the proposed framework. Functional blocks are grouped into front-end (low-level perception) and back-end (high-level perception) layers, organized by increasing computational complexity and abstraction level. Data flows are indicated by: green for raw sensor data, red for semantic information, and blue for system-level communication.

from conventional operational tools into intelligent agricultural hubs. Current agricultural scenarios often suffer from widespread infrastructure deficiencies [5], where insufficient Wi-Fi router and base station coverage hinders real-time data transmission. As dynamic aerial nodes, UAVs can establish a “device-edge-cloud” three-tier data transmission architecture [6], effectively bridging the connectivity gap between farmland and digital systems [7]. Drone swarms can execute diversified tasks, such as seeding, plant protection, and 3-D mapping, acquiring field data through multimodal sensors. After edge computing preprocessing, agricultural knowledge graphs are constructed to provide structured data support for artificial intelligence (AI) decision-making systems [8]. Their operational paradigm has transcended passive monitoring and evolved into an active agricultural intervention system that integrates precision pesticide application and intelligent harvesting.

Intelligent environmental perception forms the technological foundation for autonomous drone decision-making [9]. However, current mainstream perception solutions exhibit significant limitations: First, simultaneous localization and mapping (SLAM) technology generates only semantically deficient point cloud maps (including visual-SLAM and light detection and ranging (LiDAR)-SLAM), which are inadequate for agricultural knowledge representation. Second, object detection and image segmentation algorithms, predominantly developed on idealized 2-D datasets, fail to construct ecologically correlated 3-D growth models. Third, a deeper issue lies in existing technologies deconstructing plant organs (leaves, branches, flowers/fruits) into discrete spatial elements [10], neglecting their dynamic coupling with biological attributes and light conditions. Meanwhile, orchard environments exhibit complex textures, microscale targets, and chaotic spatial distributions, causing mainstream 3-D object estimation methods (e.g., bounding box regression from autonomous driving [11]) to maintain high error rates in fruit counting and pose estimation tasks. These challenges highlight the necessity of developing agricultural-specific perception frameworks.

To address these challenges, this article proposes a metric-semantic SLAM framework as shown in Fig. 1 for 3DSG-based plant inspection.¹ By leveraging the hierarchical structure of scene graphs, our approach significantly enhances fruit counting accuracy through error propagation control in object pose estimation. The systematic analysis of multisensor

error sources and hierarchical abstraction mechanisms represents a critical advancement in metric-semantic SLAM. The principal contributions are summarized as follows.

- 1) A 3DSG is introduced to construct a multidimensional spatiotemporal representation of plant development, facilitating interpretable growth analysis and traceable, time-series monitoring of agricultural processes.
- 2) A systematic analysis of object estimation uncertainties is conducted based on different sensor models, sensor combinations (including stereo camera, RGB-D camera, and camera-LiDAR combinations), and error propagation modeling, highlighting the critical role of hierarchical abstraction layers in enabling precision agriculture. The effectiveness of the proposed system is validated through both simulated and real-world experiments.
- 3) IoT-based smart vertical farming framework that combines mobile robotic perception systems and automated environmental control mechanisms to enable real-time data collection and closed-loop feedback regulation of crop growth metrics is proposed.

II. RELATED WORK

The research on smart and sustainable agriculture faces a dual mission: it should address the limitations of traditional agricultural production models in terms of resource efficiency and environmental adaptability, while simultaneously establishing a closed-loop scientific system to address resource-population conflicts [12]. Its technical framework comprises four core modules: 1) perception; 2) analysis; 3) decision-making; and 4) execution. The perception module, implemented through IoT networks, enables data acquisition and execution feedback, serving as the data foundation of the closed-loop system that directly determines its performance ceiling.

Despite rapid advancements in robotic perception technologies, a significant cognitive gap persists between computer algorithm paradigms and human cognition [13]: low-level perception focuses on geometric localization (e.g., spatial XYZ coordinates), while computer vision (CV)-based high-level abstract perception, despite breakthroughs in object detection, still suffers from fragmented multilevel representations in algorithms like robotic SLAM. The inherent incompatibility between heterogeneous information, such as geometric point clouds and semantic objects leads to fragmented environmental representations, ultimately constraining downstream task

¹ Simulation environment adapted from ICUAS’2024 competition, June 4-7, 2024 in Chania, Greece. Where the authors’ team achieved second place.

effectiveness like precise manipulation. To address these challenges, this article systematically reviews research progress through three dimensions: 1) sensor fusion (Section II-A) for multimodal data synergy enabling 3-D object detection; 2) metric-semantic mapping (Section II-B) bridging geometric-semantic divides; and 3) hierarchical scene graphs (Section II-C) constructing logically connected structured environmental representations.

A. Sensor Fusion for Accurate 3-D Object Detection

To establish a comprehensive environmental perception system, multiple core CV technologies, including object detection and object tracking, have been widely integrated into perception systems. Among these, 3-D object detection technology plays a foundational role by intelligently predicting the spatial coordinates, dimensional parameters, and category attributes of critical objects (e.g., crops or machinery) surrounding agricultural vehicles. This provides object-level information support for downstream tasks, such as branch pruning [14], fruit picking [15], and other labor-intensive operations. Current mainstream methodologies in this field can be categorized into three sensor-driven technical approaches: vision camera-based, LiDAR-based, and multisensor fusion solutions [16].

While current research on 3-D object detection predominantly focuses on autonomous driving and domestic service robots [11], exploration in agricultural scenarios remains in its infancy. Compared to highly structured urban environments [17], agricultural settings present three core challenges [18]: 1) complex spatial topologies formed by intertwined foliage; 2) high-density information distribution; and 3) dynamic lighting variations, which render generic perception algorithms and datasets inadequate for precision agriculture demands [19]. Existing agricultural semantic datasets exhibit significant limitations: early datasets [20] provided only 2-D RGB images lacking depth information; recent efforts like BUP20 [21] (which includes RGB-D data for greenhouse sweet pepper instance segmentation [22]) still suffer from narrow scene coverage and insufficient sample diversity. Although the Cyrill team's multimodal apple orchard dataset achieves tree-level instance segmentation [23], it fails to overcome the technical barrier of fruit-level fine-grained detection. To address these gaps, this study innovatively proposes a fruit-level object detection algorithm based on multimodal data fusion. By establishing a cross-scale feature correlation mechanism, the proposed method achieves the first synergistic optimization of spatial localization and semantic recognition for fruits in agricultural environments, effectively bridging the research void in fine-grained perception models for this domain.

B. Metric and Semantic SLAM

Through 3-D object detection algorithms, we can obtain an object's coordinates in the body frame (i.e., its position relative to the robot). However, to derive the object's world coordinates, knowledge of the robot's position relative to

the world frame is required, which necessitates SLAM technology. Metric SLAM has evolved over decades [9], driven by advancements in both hardware and software [24] [25]. Hardware innovations have spanned sensors, such as inertial measurement unit (IMU) [26], LiDAR [27], global positioning system (GPS), and multisensor fusion systems [28], [29]. Software advancements are typically divided into front-end [30] and back-end [31] components. The front-end processes raw sensor data, progressing from traditional feature detection techniques to deep learning-based feature extraction [32]. The back-end optimizes front-end outputs at lower frequencies, evolving from Kalman filters [33] to factor graph optimization [31]. Collectively, these developments have enhanced positioning accuracy and enabled metric SLAM's broad adoption across domains.

Although metric SLAM has achieved significant progress, the world, for us, is not merely a collection of geometric primitives (points [34], lines [35], surfaces [36]) or isolated pixels/LiDAR point clouds. These map elements inherently carry higher-level semantic information, motivating extensive research on integrating semantics into SLAM systems. In semantic perception, transformative progress has been made over the past decade, beginning with the ImageNet dataset and AlexNet [37]. Technologies, such as object detection and semantic segmentation [38] have matured significantly. Recently, the rise of foundation models [39] has ushered in a new era for AI, further expanding the horizons of semantic understanding.

C. 3-D Scene Graph for Hierarchical SLAM

Semantic SLAM, as an intersection of SLAM and CV, includes a wide range of studies that integrate higher-level semantic information into purely metric SLAM maps [40]. We categorize this body of work as metric-semantic SLAM, which includes various approaches [41]. However, most of these works maintain flat metric-semantic maps, where elements of different abstraction levels, such as basic landmark points and objects, are indiscriminately stored in the same map database [42]. Typically, attributes distinguish whether a point is a landmark or other object (e.g., a bottle) [43]. This fragmented combination fails to reflect the relationships between elements in the map and does not provide the robot with a holistic view of its surroundings.

To achieve a more complete and holistic perception, Armeni et al. [44] first proposed the hierarchical 3DSG to organize semantic-metric elements at different levels of abstraction. A 3DSG is a layered graph where nodes represent spatial concepts at multiple levels of abstraction (from low-level geometry to objects, places, rooms, buildings, etc.) and edges represent relations between these concepts. In this direction, Kimera [10] and Hydra et al. [13] proposed 3DSG, which is a multiresolution hierarchical graph structure to organize these semantic elements at different levels of abstraction. However, these works primarily focus on an ideal simulation setting that produces precise semantic segmentation outcomes. As reported in Hydra [13], even the ground truth (GT)-Trajectory configuration does not achieve 100% accuracy

for “% Found” and “% Correct.” Achieving entirely flawless and all-encompassing semantic segmentation results in real-world scenarios is impractical, highlighting the necessity for further improvements in object mapping accuracy under noise interference.

This article addresses the limitations of existing metric-semantic SLAM systems, which primarily focus on calculating the coordinates of objects without considering their orientation or the hierarchical relationships between different levels of abstraction. These simplifications may be acceptable in an ideal simulation environment, but are not feasible in the real world, which is characterized by noise and bias. To address these issues, this article presents a metric-semantic SLAM framework, as shown in Fig. 1, for fruit inspection based on 3DSG. By leveraging the top-down structure of the scene graph, our work significantly improves the accuracy of fruit counting by reducing the impact of various errors on object pose estimation. The systematic analysis of error sources and the hierarchical approach proposed in this article represent important advancements in the field of metric-semantic SLAM.

III. ERROR PROPAGATION ANALYSIS IN FRUIT DETECTION AND 3-D SEMANTIC MAPPING

This section systematically investigates error sources throughout the fruit perception pipeline, spanning from 2-D detection to 3-D semantic mapping in Section III-A. Then, we present error analysis of two prevalent depth sensing configurations: 1) stereo vision systems in Section III-B and 2) LiDAR-camera fusion architectures in Section III-C, with particular emphasis on systematic errors introduced during depth estimation processes.

A. Projection Model and Noise Propagation

In this part, we detail the process of recovering 3-D coordinates from 2-D image detections, propagating errors in the camera coordinate system, and transforming these errors into the world coordinate system. We use $\mathbf{R} \in \mathbb{R}^{3 \times 3}$ to denote 3-D rotation matrix, $\mathbf{t} \in \mathbb{R}^3$ to denote 3-D translation vector and $\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \in \mathbb{R}^{4 \times 4}$ to denote 3-D transformation matrix.

The camera intrinsic matrix $\mathbf{K}$ is given by

$$\mathbf{K} = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}. \quad (1)$$

1) *Recover 3-D Object From Image Plane:* Given a 2-D detection (u, v) and depth d , the coordinates in the world frame (x_w, y_w, z_w) can be recovered using the camera intrinsic matrix $\mathbf{K}$, and the transformation matrix $\mathbf{T}_{wc}$

$$\begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = \mathbf{T}_{wc} * \mathbf{K}^{-1} \begin{bmatrix} (u + n_u) \cdot (d + n_d) \\ (v + n_v) \cdot (d + n_d) \\ d + n_d \end{bmatrix} + \mathbf{n}_{\text{pose}} \quad (2)$$

where n_u, n_v represent the 2-D detection noise, n_d is the depth noise, and $\mathbf{n}_{\text{pose}}$ is the camera pose error.

The errors in the recovered 3-D coordinates can be calculated using the following formulas:

$$\begin{aligned} \sigma_{x_{\text{cam}}}^2 &= \left(\frac{d}{f_x}\right)^2 \sigma_u^2 + \left(\frac{u}{f_x}\right)^2 \sigma_d^2, & \sigma_{z_{\text{cam}}}^2 &= \sigma_d^2 \\ \sigma_{y_{\text{cam}}}^2 &= \left(\frac{d}{f_y}\right)^2 \sigma_v^2 + \left(\frac{v}{f_y}\right)^2 \sigma_d^2. \end{aligned} \quad (3)$$

The covariance matrix of the object position in the world coordinates can be calculated as

$$\Sigma_{\text{world}} = \mathbf{R} \Sigma_{\text{cam}} \mathbf{R}^T + \Sigma_{\mathbf{T}_{wc}} \quad (4)$$

where Σ_{cam} is the covariance matrix of the object position in the camera coordinates, and $\Sigma_{\mathbf{T}_{wc}}$ is the covariance matrix of the transformation $\mathbf{T}_{wc}$.

2) *2-D Detection Accuracy:* Common detection or segmentation algorithms require objects to occupy similar pixel areas in the image; otherwise, errors are likely to occur. Taking YOLO [45] as an example, the difficulties in detecting small objects at a distance are as follows.

- 1) *Resolution Limitation:* YOLO divides the input image into a grid, with each grid cell responsible for detecting one object. If the object is too small, it may fall within a single grid cell, making it difficult for the detector to accurately locate and classify it.
 - 2) *Feature Extraction:* YOLO uses convolutional neural network (CNN) to extract image features. The features of small objects may be lost or become less prominent during downsampling, leading to decreased detection accuracy.
- Therefore, when objects are at a greater distance, larger objects are detected with higher accuracy, while smaller objects are more prone to errors.

According to the law of large numbers, the average of a large number of independent random samples converges to the true expected value (mean) if it exists. In the context of depth estimation: Consider a bounding box containing pixels from the same object with similar depth. Each pixel's depth estimate can be viewed as an independent random sample. If we average the depth estimates from all these pixels, we can think of it as an approximation of the true depth. As the number of pixels (samples) increases, the average depth estimate tends to converge to the actual depth (if other factors are controlled). While for small objects, fewer pixels mean less reliable statistical sampling. Individual measurement errors have a larger impact on the overall depth estimate. The law of large numbers still applies, but with fewer samples, the convergence may be slower or less reliable.

As the number of pixels (samples) increases, the average depth estimate tends to converge to the actual depth. For small objects with fewer pixels, the individual measurement errors have a larger impact on the overall depth estimate. Assume a bounding box contains n pixels, where the depth estimate for each pixel i is Z_i , and the standard deviation of the depth estimate is σ_{Z_i} . The average depth estimate over the n pixels is: $\bar{Z} = (1/n) \sum_{i=1}^n Z_i$. According to the law of large numbers, as n increases, $\bar{Z}$ converges to the true depth Z . This is because the standard deviation of the average depth estimate decreases as $(1/\sqrt{n})$, we have

$$\sigma_{\bar{Z}} = \frac{1}{\sqrt{n}} \sqrt{\frac{1}{n-1} \sum_{i=1}^n \sigma_{Z_i}^2}. \quad (5)$$

For the case of small objects with fewer pixels, we have a smaller n . Moreover, since Z is larger for distant objects, the depth standard deviation σ_Z will be higher. With fewer pixels (n smaller), the average depth estimate $\bar{Z}$ will have a larger variance $\sigma_{\bar{Z}}$, and thus the individual measurement errors have a more significant impact on the overall depth estimate.

B. Depth Estimation Accuracy for Stereo Camera

There are many factors that affect depth error, including object distance, surface/medium characteristics, sensor technology [46], and other parameters. When the sensor combination is fixed, and both the detected object and the working environment are determined, the main influencing factor is the distance between the camera and the target. Under such situations, errors are more likely to occur when the object is either too far from the camera or too small in size. For distant targets or small objects with a small number of pixels, the main reasons for inaccurate depth estimation include detail loss, matching difficulty, and the impact of noise.

- 1) *Detail Loss*: Fewer pixels mean lower image resolution, resulting in the loss of detailed information in the image. This reduction makes it difficult for algorithms to capture the edges and textures of objects, thereby affecting the accuracy of depth estimation.
- 2) *Matching Difficulty*: Depth estimation usually relies on feature point matching in the image. Fewer pixels lead to a reduced number of feature points, thereby increasing the difficulty of matching and subsequently affecting the accuracy of depth estimation.
- 3) *Noise Impact*: As pixel-per-meter decreases, meaning each pixel corresponds to a larger actual spatial range, the noise accumulated within that space also increases, resulting in a lower signal-to-noise ratio. Low-resolution images are more susceptible to noise, which can interfere with the normal operation of depth estimation algorithms, leading to inaccurate estimation results.

Taking the stereo camera as an example, based on the disparity-depth model, Mallick et al. [46] calculated the depth from disparity and determined the accuracy and resolution of depth. Using similar triangles, the depth $Z(\rho)$ of an object can be described by

$$Z(\rho) = \left(\frac{m}{fb} \rho + \left(\frac{1}{Z_r} + \frac{n}{fb} \right) \right)^{-1} \quad (6)$$

where m and n are the parameters of linear normalization, f is camera focal length, b is the stereo baseline length, and ρ is the normalized disparity [46]. To determine the standard deviation σ_Z , the derivative of $Z(\rho)$ is initially computed with respect to ρ

$$\begin{aligned} \frac{dZ}{d\rho} &= \frac{d}{d\rho} \left(\left(\frac{m}{fb} \right) \rho + \left(\frac{1}{Z_r} + \frac{n}{fb} \right) \right)^{-1} \\ &= - \left(\frac{m}{fb} \right) \left(\left(\frac{m}{fb} \right) \rho + \left(\frac{1}{Z_r} + \frac{n}{fb} \right) \right)^{-2}. \end{aligned} \quad (7)$$

Upon substituting $Z(\rho)$ into (7), we obtain $(dZ/d\rho) = -(m/[fb])Z^{-2}$. The standard deviation σ_Z is related to the standard deviation of the normalized disparity σ_ρ can be given by

$$\sigma_Z = \left| \frac{dZ}{d\rho} \right| \sigma_\rho = \left| - \left(\frac{m}{fb} \right) \frac{1}{Z^2} \right| \sigma_\rho = \left(\frac{m}{fb} \right) \frac{\sigma_\rho}{Z^2}. \quad (8)$$

From the above expression, the error of depth measurement is proportional to the square of the depth. This has been experimentally verified in [46].

Assuming the errors in the estimation of X and Y coordinates are independent, one has

$$\sigma_X = \left(\frac{mx}{f^2b} \right) Z^2 \sigma_\rho \quad \text{and} \quad \sigma_Y = \left(\frac{my}{f^2b} \right) Z^2 \sigma_\rho \quad (9)$$

where x and y are the coordinates of the pixel in the depth image, and σ_X and σ_Y are the errors in X and Y , respectively. Both the depth accuracy and resolution decrease quadratically with increasing depth from the sensor.

C. Depth Estimation Accuracy for LiDAR Camera Fusion

The LiDAR measures a point in 3-D space using a bearing direction $\omega_i \in \mathbb{S}^2$ and depth d_i . The bearing direction in terms of azimuth θ and elevation ϕ angles, the bearing direction ω_i is parameterized as

$$\omega_i = [\cos \theta \cos \phi \quad \sin \theta \cos \phi \quad \sin \phi]. \quad (10)$$

The measured point in the LiDAR frame is denoted as $\mathbf{P}_i = d_i \omega_i$. The uncertainty in the LiDAR measurement arises from the following two sources.

- 1) *Bearing Direction Noise*: $\delta_{\omega_i} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\omega_i})$ in the tangent plane of ω_i .
- 2) *Depth Noise*: $\delta_{d_i} \sim \mathcal{N}(0, \Sigma_{d_i})$ in the depth measurement.

The true point $\mathbf{P}_i^{\text{gt}}$ relates to the measured point as

$$\mathbf{P}_i^{\text{gt}} = (d_i + \delta_{d_i})(\omega_i \oplus_{\mathbb{S}^2} \delta_{\omega_i}) \approx d_i \omega_i + \omega_i \delta_{d_i} - d_i \omega_i^\wedge \mathbf{N}(\omega_i) \delta_{\omega_i} \quad (11)$$

where $\mathbf{N}(\omega_i)$ is an orthonormal basis for the tangent plane, and $\omega_i^\wedge$ is the skew-symmetric matrix of ω_i .

The total noise in $\mathbf{P}_i$ is calculated as

$$\mathbf{w}_i = \omega_i \delta_{d_i} - d_i \omega_i^\wedge \mathbf{N}(\omega_i) \delta_{\omega_i} \sim \mathcal{N}(\mathbf{0}, \Sigma_{\mathbf{P}_i}) \quad (12)$$

with covariance

$$\Sigma_{\mathbf{P}_i} = \omega_i \Sigma_{d_i} \omega_i^T + d_i^2 \omega_i^\wedge \mathbf{N}(\omega_i) \Sigma_{\omega_i} \mathbf{N}(\omega_i)^T \omega_i^{\wedge T}. \quad (13)$$

The LiDAR point $\mathbf{P}_i$ is transformed to the camera frame using extrinsic calibration $\mathbf{T} = (\mathbf{R}, \mathbf{t})$

$$\mathbf{P}_i^c = \mathbf{R} \mathbf{P}_i + \mathbf{t}. \quad (14)$$

The uncertainty in $\mathbf{P}_i^c$ is calculated by

$$\Sigma_{\mathbf{P}_i^c} = \mathbf{R} \Sigma_{\mathbf{P}_i} \mathbf{R}^T. \quad (15)$$

Then, the point $\mathbf{P}_i^c = (X_i^c, Y_i^c, Z_i^c)$ in the camera frame is projected onto the image plane via the following formula:

$$\mathbf{p}_i = \left(f_x \frac{X_i^c}{Z_i^c} + c_x, f_y \frac{Y_i^c}{Z_i^c} + c_y \right) \quad (16)$$

where f_x and f_y are focal lengths and (c_x, c_y) is the principal point.

The Jacobian of Z_i^c with respect to $\mathbf{P}_i$ is computed by

$$\frac{\partial Z_i^c}{\partial \mathbf{P}_i} = \mathbf{R}_3 \quad (17)$$

where $\mathbf{R}_3$ is the third row of $\mathbf{R}$. The variance of Z_i^c is

$$\sigma_{Z_i^c}^2 = \mathbf{R}_3 \Sigma_{\mathbf{P}_i} \mathbf{R}_3^T. \quad (18)$$

Assuming isotropic bearing noise ($\Sigma_{\omega_i} = \sigma_{\omega_i}^2 \mathbf{I}_{2 \times 2}$)

$$\Sigma_{\mathbf{P}_i} = \sigma_{d_i}^2 \omega_i \omega_i^T + d_i^2 \sigma_{\omega_i}^2 (\mathbf{I}_{3 \times 3} - \omega_i \omega_i^T). \quad (19)$$

The depth uncertainty becomes

$$\sigma_{Z_i^c}^2 = \sigma_{d_i}^2 (\mathbf{R}_3 \omega_i)^2 + d_i^2 \sigma_{\omega_i}^2 (\|\mathbf{R}_3\|^2 - (\mathbf{R}_3 \omega_i)^2). \quad (20)$$

Note the term $\sigma_{d_i}^2 (\mathbf{R}_3 \omega_i)^2$ is generally negligible in magnitude, while the dominant contribution comes from $d_i^2 \sigma_{\omega_i}^2 (\|\mathbf{R}_3\|^2 - (\mathbf{R}_3 \omega_i)^2)$. This predominant term exhibits quadratic growth trend with respect to the distance d_i , which makes the second term in (20) becoming increasingly significant as d_i increases. This observation is well aligned with the depth uncertainty analysis results in Section III-B, and we will introduce hierarchical scene graph to alleviate the uncertainty introduced by small or distant objects in Section IV.

IV. SCENE GRAPH

Hierarchical abstraction intrinsically mitigates uncertainty through two key mechanisms.

- 1) *Error Propagation Mitigation*: Noisy detections (e.g., small fruits) are confined to lower abstraction levels, preventing error propagation to the plant/orchard.
- 2) *Redundancy Exploitation*: Higher-level objects (e.g., plants) aggregate observations from multiple child nodes, suppressing outliers through statistical averaging.

In this section, we first analyze absolute/relative errors and abstraction-level implications to justify our hierarchical semantic mapping approach in Section IV-A, then we formalize our 3DSG framework in Section IV-B.

A. Absolute and Relative Errors in Hierarchical Structures

In semantic SLAM, objects are typically simplified to only retain their category, size, and pose information. Thus, each object carries information in the form of $[class, size, T \in \mathbb{R}^{4 \times 4}]$, which is the information we need to extract in metric-SLAM. However, the number of pixels generated by each object in the image is different, as shown in Fig. 2, meaning that the amount of information in the image (the source of information) will also be different. We are mainly concerned with the uncertainty of object position and size.

According to (3), the accuracy of the position is significantly affected by the uv and $depth$ errors. The uncertainty of uv and

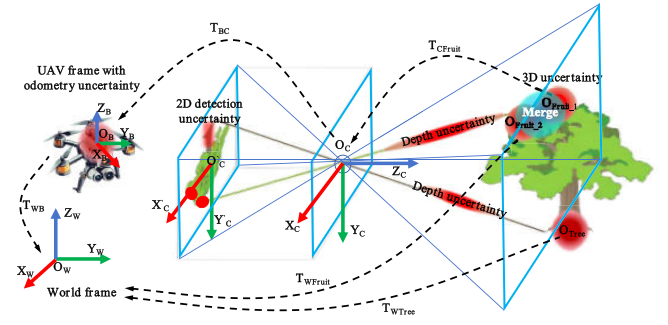


Fig. 2. Projection model and uncertainty propagation.

depth accuracy is influenced by the size of the object in the image, which is affected by the size and distance of the object. This relationship can be expressed by the following formula:

$$\text{obj_image_size} = \frac{\text{obj_size} \cdot f}{\text{distance}} \quad (21)$$

where f is the camera focal length, which can be obtained from the camera intrinsic matrix $\mathbf{K}$. Regarding uv and depth errors, they are affected by the size of the object in the image. In particular, it is necessary to ensure that the object image size reaches a suitable size. In the following, we analyze the relationship between 2-D detection errors, depth errors, and object image size.

The localization error of an object can be divided into absolute error and relative error. The absolute error xyz_error is measured in a unified length unit (m). The relative error is given by $(xyz_error/\text{size}_{obj})$. A natural assumption in object association is that two objects of the same class cannot overlap in 3-D space, so the distance threshold between objects is usually set to the size of the object itself, and the size values are determined through statistical evaluation of object distributions. Large objects have larger distance thresholds, while small objects have smaller distance thresholds. Therefore, it is more reasonable and accurate to first associate the larger objects and then the smaller objects within them.

From another perspective, in the real world, larger objects logically contain (or correspond to) multiple smaller objects (for example, a tree has multiple fruits, and a building has multiple rooms). Smaller objects are less likely to correspond to multiple larger objects (for example, a fruit is not on multiple trees, or a room does not belong to multiple buildings). Therefore, a hierarchical scene graph is a natural organizational structure, allowing the metric-semantic SLAM framework to organize semantic elements at different levels of abstraction. This approach also reduces the memory required for large-scale mapping. More importantly, larger objects are associated with more depth pixels, making the depth values less susceptible to outliers and more robust.

B. Hierarchical Scene Graph for Fruit Inspection

We use a 3DSG to organize objects at different levels of abstraction. A 3DSG is a layered graph with nodes representing spatial concepts at multiple levels of abstraction (from low-level geometric point clouds to objects, places, and rooms for indoor service robots).

1) *Scene Graph Definition for Plant Inspection*: In our case, to achieve the goal of managing the orchard, the abstraction levels rise from purely geometric points without semantic information to fruits and so on, with the highest level being the orchard. The stems and leaves of the plants should be included in the hierarchy. To simplify the problem, this study only considers point clouds, fruits, plants, and orchards. The relationships between nodes include subordination and adjacency. The formal definition of our 3DSG is shown as follows.

Definition 1 (Hierarchical Graph): A graph $G = (V, E)$ is said to be hierarchical if the set of nodes V can be partitioned into l layers, i.e., $V = \bigcup_{i=1}^l V_i$, such that:

- 1) *Single Parent*: Each node $v \in V_i$ at layer i shares an edge with at most one node in the layer V_{i+1} ;
- 2) *Inherent Position Attributes*: Each node $v \in V_i$ at layer i pertains the position attributes from its parent node in the layer V_{i+1} ;
- 3) *Locality*: Each node $v \in V_i$ can only share edges with nodes in the same or adjacent layers, i.e., V_{i-1} , V_i , or V_{i+1} ;
- 4) *Disjoint Children*: For any $u, v \in V_i$, the children of u and v , namely $C(u)$ and $C(v)$, are disjoint (i.e., they share no nodes or edges), where

$$C(u) = \{w \in V_{i-1} \mid (w, u) \in E\} \quad (22)$$

denotes the children of u , i.e., the set of nodes in layer V_{i-1} sharing an edge with u . We refer to G as an L -layered hierarchical graph and denote with V_l the set of nodes in layer l . Moreover, we conventionally order the layers from the bottom to the top, i.e., the lowest layer is V_1 , while the top layer is V_L .

For example, in the case of fruits, the highest layer of the hierarchical scene graph is the orchard, the next layer is the plants, the third layer is the fruits, and the lowest layer is purely geometric landmarks without semantic meaning. Each fruit belongs to a plant, and each plant belongs to an orchard.

2) *Pose Estimation and Pose Update With Parent*: This study develops an updated strategy that allows child nodes and parent nodes to update their poses together. First, the 2-D detection algorithm detects objects at lower abstraction levels only within the bounding box of objects at higher abstraction levels. For example, fruits are detected only within the bounding box of a plant, and the fruit node is directly added under the plant node by Algorithm 1: *add_node(fruit)*. In this logic, larger objects are associated with more depth pixels, making the corresponding depth estimation less susceptible to outliers and more robust. These objects have higher abstraction levels in the scene graph. The algorithm queries objects at higher abstraction levels first, then queries objects at lower abstraction levels sequentially. When the position of an object is updated, the positions of its child nodes are also updated, which helps reduce matching errors during loop closure.

Algorithm 1 sets different distance thresholds based on the abstraction level and actual size of the object. This algorithm checks the list of child nodes: if the current node has no child nodes, the new child node is directly added. If the current node has child nodes, the algorithm iterates through each

Algorithm 1 Add Child to Scene Graph

```

1: if self.children is empty then
2:   add_child()
3: else
4:   for each old_child in self.children do
5:     node_dist = calc_node_dist(old_child, child)
6:     if node_dist < dist_threshold then
7:       merge_existing_child_with_child()
8:       return existing_child
9:   else
10:    pass
11:   end if
12: end for
13: add_child()
14: end if
15: update_node_attributes()
16: return child

```

child node and calculates the distance (node_dist) between the new child node and the existing child nodes. If the calculated distance is less than the threshold, the new child node is merged with the corresponding existing child node using the infinite impulse response (IIR) filter. Subsequently, the system updates the pose of all children of this merged node. If no mergeable child node is found, the new child node is appended to the list of child nodes. By querying and updating layer by layer, the algorithm effectively manages and maintains the relationships between objects in the scene graph. The framework is inherently extensible. Adding abstraction levels would require: 1) modifying the 3-DSG class hierarchy to include new semantic levels and 2) retraining detection models with annotated plant-bed data to enable recursive graph construction.

V. EXPERIMENT

As analyzed in Section III, we conducted experiments using two different sensor configurations: 1) RGB-D and 2) camera-LiDAR fusion. Four different datasets collected in different experimental scenarios were tested. The detailed configurations and parameters for each scenario are presented in Table I. The quantitative evaluation in this section compares algorithmic fruit counts against GT across multiple fruit varieties. To provide an intuitive measure of counting accuracy, we include bracketed percentages indicating each algorithm's closeness to GT, where 100% represents perfect alignment with GT. Deviations above or below this value reflect performance degradation, with greater differences indicating lower accuracy.

For RGB-D datasets, the state-of-art metri-semantic SLAM, OA-SLAM [41], Quadric-SLAM [40], VOOM [47], and 3DSG SLAM, Kimera [10], and Hydra [13] are selected as the main baseline. OA-SLAM, Quadric-SLAM and VOOM use dual quadric as landmarks in SLAM with constraints of detected bounding boxes or instance segmentations, while all three above methods apply flat metric-semantic maps for point cloud landmarks and objects (represented by quadrics)

TABLE I
EXPERIMENT DATASETS CONFIGURATION

Dataset	Sensor	Platform	Scenario	Figure	Object	Object Color	BG Color	Quantity
Gazebo	RGB-D	UAV	Vertical farm	Fig. 1	Plant fruit	Red, yellow, purple	Oak	50-100
TUM-RGBD	RGB-D	Handheld	Indoor	Fig. 4	Desktop stuff	Multi color	Gray	50-60
ICUAS'24	Camera-LiDAR	UAV	Outdoor orchard	Fig. 5	Fake fruits	Red, yellow	Green	20-30
Hydroponic	Camera-LiDAR	UAV	Hydroponics	Fig. 8	Strawberry	Green, red	Green	10-50

storage, i.e., they are not concerned with estimating higher-level semantics and store all mapping elements at the same level of abstraction. Kimera and Hydra applied 3DSG to organize semantic elements at different levels of abstraction. Their studies, however, mainly considered an ideal simulation environment with accurate semantic segmentation results. It is unrealistic to obtain completely correct and comprehensive semantic segmentation in the real world. Therefore, it remains crucial to improve object mapping accuracy under the influence of noise.

For LiDAR-camera fusion datasets, Kang et al. [48] and Liu et al. [49] methods are selected as the main benchmarks. Among them, Kang's method is only camera-LiDAR fusion without odometry information, so it can only estimate the fruit position statically in the local frame (the vehicle cannot move, only the manipulator can move to grasp fruits), and cannot calculate the fruit position in the global frame. Liu's method extends Kang's method, but with worse positioning accuracy. Both methods project the LiDAR point cloud onto the image plane to obtain the depth of pixels in the image, and then estimate the 3-D position of the fruit based on the depth. In contrast, our sensor fusion method adopts the 3-D object detection paradigm commonly used in autonomous driving. We project the object detection bounding box from the image into 3-D space, and subsequently filter the point clusters in the 3-D space to avoid the influence of noisy point clouds. To address potential calibration challenges from variable extrinsic settings, we employ a map-based fusion strategy where we first prebuild a reference point cloud map using LiDAR odometry, then perform camera-LiDAR fusion exclusively on this prebuilt map. This approach enables higher positioning accuracy and more robust performance.

A. Gazebo Simulation for Fruit Counting

In the Gazebo environment as depicted in Fig. 1, various fruits are placed on the vertical plant bed in the garden. Each bed contains one of the three species of plants. Each type of plant has different colors and sizes, namely purple eggplants, red tomatoes, and yellow peppers. A UAV flies in the greenhouse to conduct fruit counting, a depth camera, an IMU are installed on the simulated UAV. The GT odometry information is used to provide the pose information of the UAV.

We conducted a comparative analysis of fruit counting accuracy with Kimera and Hydra frameworks, given their shared experimental configuration utilizing RGB-D camera inputs and ground-truth semantic segmentation annotations. Within an idealized simulation framework, precise semantic segmentation labels can be systematically generated through HSV-based thresholding techniques, effectively bypassing sensor noise

TABLE II
FRUIT COUNTING ACCURACY IN GAZEBO SIMULATION SETUP
PROVIDED BY ICUAS'24 SIMULATION PHASE

	Kimera	Hydra	Ours	GT
Eggplant	27 (122%)	24 (109%)	22 (100%)	22
Pepper	49 (114%)	38 (93%)	42 (102%)	41
Tomato	39 (114%)	32 (95%)	34 (100%)	34
Plant	21 (110%)	19 (100%)	19 (100%)	19

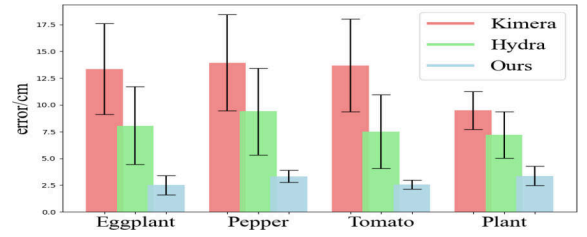


Fig. 3. Plant and fruit reconstruction position error comparison in centimeters.

artifacts. As shown in Table II, our Gazebo simulation results demonstrate exceptional counting precision: the proposed methodology achieves exact 100% alignment with GT in most test cases, with a single instance at 102% - still well within near-perfect range. This significant performance improvement over comparative methods validates our approach's effectiveness in fruit counting applications.

It is notable that even with the GT-Trajectory configuration, achieving 100% accuracy for %Found and %Correct in the simulation scenes using ground-truth semantics is not realized. There are centimeter-level fruit and plant localization errors in all three methods as shown in Fig. 3. Two primary factors contribute to the imperfect performance of the GT-Trajectory setup. First, Hydra's object extraction method tends to under-segment nearby objects of the same class, resulting in the predicted object centroid deviating further from the individual GT object instances. Second, the underlying metric-semantic reconstruction within the active window (such as Kimera and Voxblox) can occasionally cause object labels to be projected onto surfaces behind objects, leading to the generation of hallucinated objects. To address these issues, the algorithm design has been refined to be more robust to noise and uncertainty, following the analysis outlined in Section III.

B. TUM RGB-D Dataset

Since benchmark methods are designed for service robotic applications rather than being specifically targeted toward fruit mapping and counting, we validate the generalizability of

TABLE III
OBJECT COUNTING ACCURACY IN TUM RGB-D
DATASET AND DIAMOND DATASET

	OA	QISO	VOOM	Ours	GT
Fr2_desk	119 (270%)	64 (145%)	49 (111%)	46 (105%)	44
Fr2_person	126 (280%)	69 (153%)	56 (124%)	49 (109%)	45
Fr3_office	-	-	66 (108%)	58 (95%)	61
Diamond_1	73 (146%)	59 (118%)	53 (106%)	48 (96%)	50
Diamond_2	74 (172%)	78 (181%)	50 (116%)	47 (109%)	43

our proposed approach beyond agricultural scenarios through testing on the TUM RGB-D dataset, which primarily contains indoor environments. As discussed in OA-SLAM [41], the mere incorporation of bounding boxes or quadric surface models as map landmarks may not substantially enhance SLAM's localization accuracy. Consequently, rather than prioritizing localization precision comparisons, our evaluation framework emphasizes assessing and comparing the quantitative reconstruction completeness of objects through semantic-aware metrics in SLAM implementations.

This evaluation by counting object numbers reflects the accuracy of object-level mapping. The closer the number of reconstructions is to the GT, the better the performance of data association. Since the TUM RGB-D dataset does not provide GT 3-D reconstruction or the GT counts and labels of objects, we obtain the GT object counts by manual labeling and visual inspection.

Table III summarizes the results. We compare our algorithm with three object data association methods: 1) OA-SLAM; 2) QISO-SLAM; and 3) VOOM. Among them, OA-SLAM and QISO-SLAM, IoU serves as the primary metric for object association. OA-SLAM restricts associations to objects of the same category, whereas QISO-SLAM associates objects based solely on IoU values, without considering their categories, and VOOM applies Wasserstein distance [47] for object association.

Comparatively, our method outperforms other algorithms on all sequences by scrutinizing the object's class, size, and 3-D position in the scene graph, allowing for the updating of object poses. Therefore, our method performs well even when dealing with sequences like Fr2_person, where moving persons are present, while VOOM tends to yield a higher number of incorrect counts. The sparse 3DSG reconstruction result of the TUM RGB-D dataset is shown in Fig. 4.

The results for OA-SLAM and QISO-SLAM are consistent with those reported in [47], as the source code for QISO-SLAM remains unavailable. Other results represent the best performance achieved. It can be observed that OA-SLAM exhibits the highest number of redundant objects among all the methods. This outcome is attributed to its utilization of 2-D IoU alone for object association, which leads to erroneous object detection, misclassification, and consequently, erroneous generation of 3-D objects. And VOOM outperforms QISO-SLAM by incorporating additional criteria for object data association.

C. ICUAS'24 Competition Dataset

The ICUAS'24 competition is organized by LARICS from the University of Zagreb. Inspired by the scenario of robots

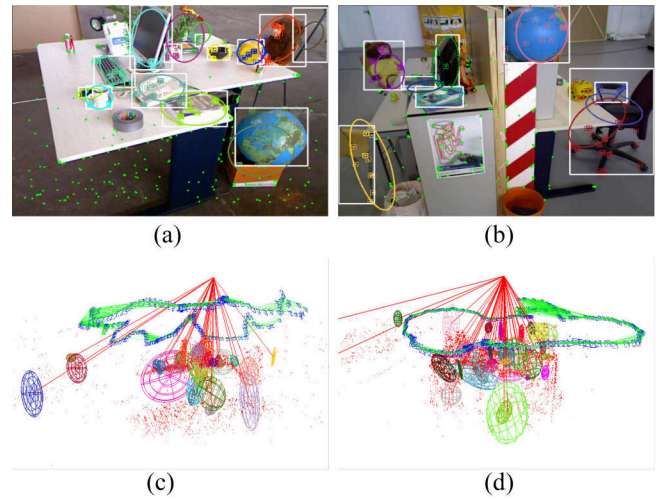


Fig. 4. TUM RGB-D dataset test results. (a) 2-D object detection in fr2_office. (b) 2-D object detection in fr3_office. (c) 3DSG in fr2_office. (d) 3DSG in fr3_office.



Fig. 5. ICUAS'24 dataset collection scenario. (a) Dataset collection orchard environment. (b) UAV configuration.

in agriculture, the final stage of the competition takes place in an orchard shown in Fig. 5(a). There are two different types of plants in the orchard, with differently colored fruits (yellow and red). A UAV is deployed and flown manually, both in position and attitude modes, to collect data (including LiDAR point cloud, camera image, GPS signal, and IMU data) for fruit counting. The UAV is equipped with a front-facing camera, slightly pitched-down 3-D lidar, GPS receiver, IMU, and Pixhawk controller as shown in Fig. 5(b).

Kang's method and Liu's method primarily focus on fruit localization. However, the ICUAS'24 competition did not provide GT 3-D fruit locations (in fact, obtaining GT object localization in outdoor, unstructured, and cluttered environments like orchards is very challenging). Therefore, we only compared our method with Liu's method for fruit counting accuracy. Kang's method uses a Livox LiDAR (a solid-state LiDAR with a nonrepetitive scan pattern) to accumulate point clouds from one static pose and project them onto the image plane to estimate pixel depth. This method is only suitable for static platforms or robots that need to stay in the same position for a long time, which significantly reduces the robot's efficiency (both in detection and fruit harvesting). Consequently, it is not suitable for mobile robots, especially UAVs. Liu's method builds on Kang's method by adopting a sliding window to accumulate point clouds from the LiDAR and then projecting the LiDAR point clouds onto the image plane.

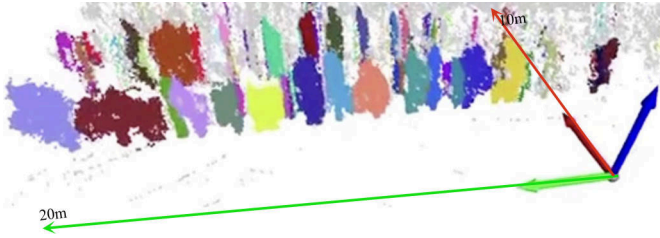


Fig. 6. Tree Segmentation in the Orchard.

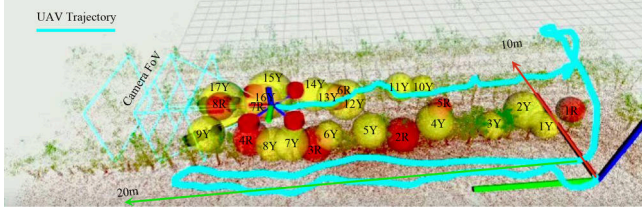


Fig. 7. ICUAS'24 orchard mapping results with fruit landmarks and point cloud mapping.

TABLE IV
OBJECT COUNTING ACCURACY IN ICUAS'24 DATASET

	Liu's(Y)	Ours(Y)	Liu's(R)	Ours(R)	GT(Y R)
Bag 1	21 (117%)	17 (94%)	12 (150%)	8 (100%)	18Y 8R
Bag 2	26 (144%)	21 (117%)	11 (138%)	10 (125%)	18Y 8R
Bag 3	27 (150%)	18 (100%)	9 (113%)	7 (88%)	18Y 8R
Bag 4	29 (126%)	22 (96%)	19 (119%)	14 (88%)	23Y 16R

However, due to parallax and occlusion, in an actual orchard, the background depth may be projected onto the same pixels as the foreground depth, which introduces significant errors. Our method directly selects candidate point cloud clusters in 3-D space and then probabilistically selects the fruit depth. By using IIR filters and hierarchical scene graphs to select and update the fruit candidate poses, our approach effectively avoids a large number of erroneous estimates. The tree segmentation in the point cloud map is carried out by using the DB-Scan algorithm, as shown in Fig. 6.

Since Liu's algorithm is not open-source, we reimplemented their methods based on the descriptions provided in their article and tested them on the ICUAS'24 dataset. In this dataset, the UAV flies over two rows of orchards, and the fruit mapping results can be seen in Fig. 7. In our method, all fruits are well aligned with the trees, while in Liu's method, due to background depth noise, many fruits erroneously fall into the empty area between the two rows of orchards. Quantitative results are shown in Table IV, where it can be seen that our method achieves higher accuracy compared to Liu's method. However, some errors persist, primarily due to the cluttered environment and the inherent challenges associated with fruit counting in complex agricultural scenes.

D. Hydroponic Laboratory Test for Strawberry Mapping

We also conducted indoor strawberry hydroponic cultivation at the Hong Kong Polytechnic University and performed UAV flight tests in the vertical hydroponics farm lab to

collect data and test our fruit counting system. As shown in Fig. 8, our hydroponic laboratory is an indoor controlled environment, capable of adjusting growth conditions through various equipment (including temperature, humidity, nutrient supply, CO_2 concentration, and lighting), thereby providing a tailored and stable growth environment for crops. On one hand, indoor cultivation is not limited by climate, allowing crops that are typically grown in winter to be cultivated in summer, thus ensuring continuous and stable year-round production. On the other hand, due to the scarcity of arable land in Hong Kong, indoor controlled hydroponics offers more flexible site selection compared to traditional outdoor greenhouse hydroponics, since they can be set up in factory premises, containers, and other locations, thereby minimizing environmental impact.

Our indoor controlled hydroponic platform (maintained at a temperature of $22 \pm 1^\circ C$ and relative humidity of 50%–60%) currently has 32 intelligent hydroponic units that can control light intensity, spectrum, and water circulation for research purposes. The hydroponic platform already has many sensors to monitor the plant growth environment, but most of them are for monitoring temperature, humidity, and nutrient composition, lacking sensors for monitoring plant growth status. This limitation is partly due to the structural complexity of the hydroponic platform, which makes the installation of cameras and other sensors challenging, and partly because densely installing cameras on a large-scale hydroponic platform would significantly increase costs. By equipping a UAV with an RGB-D camera and a Livox MID-360 LiDAR, a mobile data collection platform for crop monitoring is established, substantially reducing sensor installation costs. The sensor setup of the UAV could be found at [16]. In the future, manipulators can be installed on UAVs for active operations, such as pollination, pruning, and harvesting.

First, we collected over 1200 images for strawberry annotation and classification, only annotating strawberries with fully fallen petals. Based on color (green and red), the strawberries were classified into two categories: 1) ripe and 2) unripe. Unripe strawberries are difficult to be accurately identified, and the training process may incorrectly identify ripe strawberries as unripe. In contrast, fully ripe strawberries are relatively easier to be identified correctly. Additionally, manual selections may result in different classifications due to varying levels of experience, posing further challenges for data labeling. In this study, strawberries exhibiting more than 60% red surface area were classified as ripe, while those with less than 60% red area were classified as unripe. The annotated dataset was subsequently used to train a YOLOv8 network for real-time strawberry detection. The detection results are shown in Fig. 9. The detected bounding boxes are filtered and then input into the hierarchical scene graph for 3-D semantic mapping.

The final mapping of strawberries in the indoor hydroponic platform is presented in Fig. 10. It can be seen in the figure that most ripe strawberries were detected and correctly mapped in 3-D space. Since unripe strawberry detection is more challenging, only ripe strawberries are counted for final comparison. The strawberry counting results are shown in

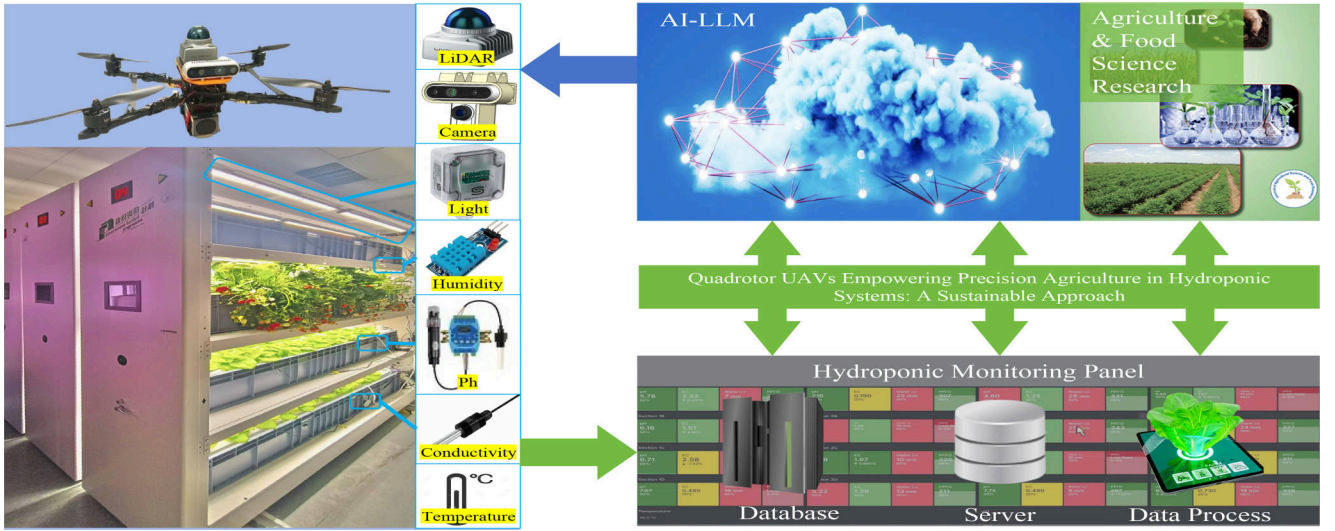


Fig. 8. System architecture of the intelligent hydroponic laboratory with embedded sensors and UAV.



Fig. 9. Ripe and unripe strawberry detection by YOLOv8. (a) Scene 1. (b) Scene 2.

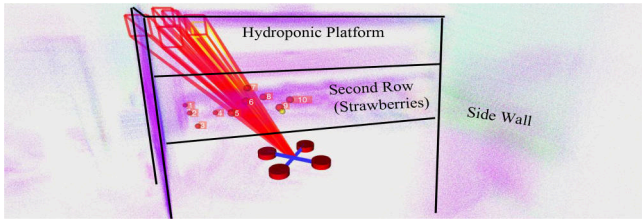


Fig. 10. 3-D strawberry mapping in the hydroponic platform.

TABLE V
STRAWBERRY COUNTING ACCURACY IN HYDROPONIC PLATFORM

	Liu's [49]	Kang's [48]	Ours	GT
Sequence 1	15 (150%)	9 (90%)	10 (100%)	10
Sequence 2	29 (132%)	17 (77%)	26 (118%)	22
Sequence 3	52 (141%)	29 (78%)	45 (122%)	37

Table V. Accurately detecting unripe strawberries will be our future work direction. For quantitative assessment, ground-truth positions of strawberries were manually measured, with the hydroponic shelf designated as the coordinate origin. In dynamic scenarios, our method achieves 2 cm localization error, which is better than Liu's method as shown in Fig. 11. Based on the above analysis, the proposed algorithm helps to achieve superior accuracy, clearly demonstrating the effectiveness of our approach.

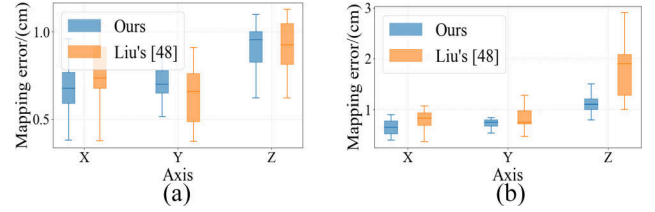


Fig. 11. Hydroponic platform strawberry 3-D mapping accuracy comparison. (a) Static mapping accuracy. (b) Dynamic mapping accuracy.

VI. CONCLUSION

This article, grounded in fruit counting, introduces a hierarchical metric-semantic SLAM system based on 3DSG for object mapping and counting. By leveraging a hierarchical scene-graph structure, the system mitigates mapping errors and significantly enhances fruit counting accuracy. Our work focuses on two key aspects: 1) reducing errors in each stage (2-D detection, depth estimation, robot odometry) and 2) leveraging a hierarchical structure in the real world to minimize object mapping errors through a top-down data association paradigm. Experimental results across four different scenarios demonstrate both the mapping accuracy of 3-D objects and the generality of the proposed algorithm. Experimental evaluations reveal context-dependent performance:

- 1) *In Outdoor Orchards (Section V-C)*: Our method reduced fruit localization errors by 40% compared to Liu's [49] method through 3-D clustering, but remained sensitive to extreme foliage density.
- 2) *In Indoor Vertical Farms (Section V-D)*: Our method achieved 92% strawberry counting accuracy (dynamic flights) by leveraging hierarchical constraints to overcome occlusion.

Fundamental limitations include: 1) degraded RGB-D performance under challenging scenarios (addressed via LiDAR fusion) and 2) heavy computational load on embedded hardware. In the future, we will optimize real-time inference via lightweight graph pruning.

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